

PRESENTATION ON

RANA PRATAP SAGAR DAM

WATER GUARDIANS

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INTRODUCTION

RANA PRATAP SAGAR DAM

The Rana Pratap Sagar Dam is a prominent hydro-engineering structure built on the Chambal River in the Chittorgarh district of Rajasthan, India. Stands as the second dam developed under the Chambal Valley Development Project. It is a key hydro-

historical Background

- o It is located on the Chambal River in Chittorgarh district, Rajasthan. And was completed in 1970, It took 3.5 years to complete (pre construction + main construction)
- o The cost estimation is upto ₹ 10,356.68 crore
- o The dam serves multiple purposes, including irrigation, generation of hydroelectric power, flood management, and water storage.

Long-term

Impact

- o Transformed arid regions through consistent irrigation
- o Brought electricity access to both rural and urban areas
- o Symbolizes inter-state cooperation and engineering progress
- o Continues to support agriculture, industry, and daily life in central India

Dam Type and

Specifications

Type: Gravity masonry dam

A gravity dam takes help of gravity to hold water (basically its own weight)

It's built strong and heavy so that water can't push it over.

Operational System:

Functions as part of a coordinated grid, dependent on upstream water management

Technical

specifications:

Total length:- 1,143 meters (3,750 ft).

Total Height:- 53.8 meters (177 ft)

Total storage:- 2.858 TMC (Thousand Million Cubic Feet) by both the reservoirs

Land required:- 1376.44 ha

(871.4 ha forest, 505.04 ha

Reservoirs

- **Upper Reservoir:**

- Live Storage: 1.42 TMC
- Max Height: 52 m
- Top width: 10 m

- **Lower Reservoir:**

- Live Storage: 1.43 TMC
- Max Height: 20 m
- Length: 7860 m



- **Evaporation Losses:** Estimated at 10–12% of average live storage per year, to be replenished from the main reservoir or rainfall.

Why the project was conceived

The ~~Pana Pratap Sagar~~ Dam project was conceived as part of India's post-independence efforts to utilize its water resources for **irrigation, power generation, and regional development**, especially in the semi-arid regions of **Rajasthan**.

Main reasons were:

1

Rajasthan had to face less and uneven rainfall. Also farmers were dependent on monsoons which lead to draught and crop failures.

2

Rajasthan had limited power infrastructure and fast flowing Chambal had high potential for clean hydroelectric energy

3

The dam was the second stage of the larger Chambal Valley Development Project, this project was mainly to control floods , harness river water and provide integrated waster resource management with a collaborative effort between Rajasthan and Madhya P. And many more...

Power

Generation

Generation is the Core

Objective of Rana Pratap Sagar Dam

- The Power generated is equally shared between Rajasthan and Madhya Pradesh
- The Chambal River, with its steep gradient and strong flow, offered ideal conditions for hydroelectric generation.
- Hydroelectric plant shows low operating cost and produce renewable , non-polluting energy unlike thermal plants

Water released from the dam flows into penstocks
(large steel pipes) which direct it to the turbines.

The Rana Pratap Sagar Hydroelectric Power House is uniquely designed as an underground facility, built near Rawatbhata to efficiently harness hydroelectric power.

- It is equipped with:
 - 4 Kaplan turbines
 - penstock diameter is 6.096m
 - Each turbine has a capacity of **43 MW** with **Total installed capacity of 172 MW**
 - **Transformers:** 24 single-phase, 18 kV/400 kV, 102 MVA

That's enough to power over 1,00,000 average Indian homes (depending on usage).

Kaplan Turbines are highly efficient for moderate water head and large flow rates.

Power

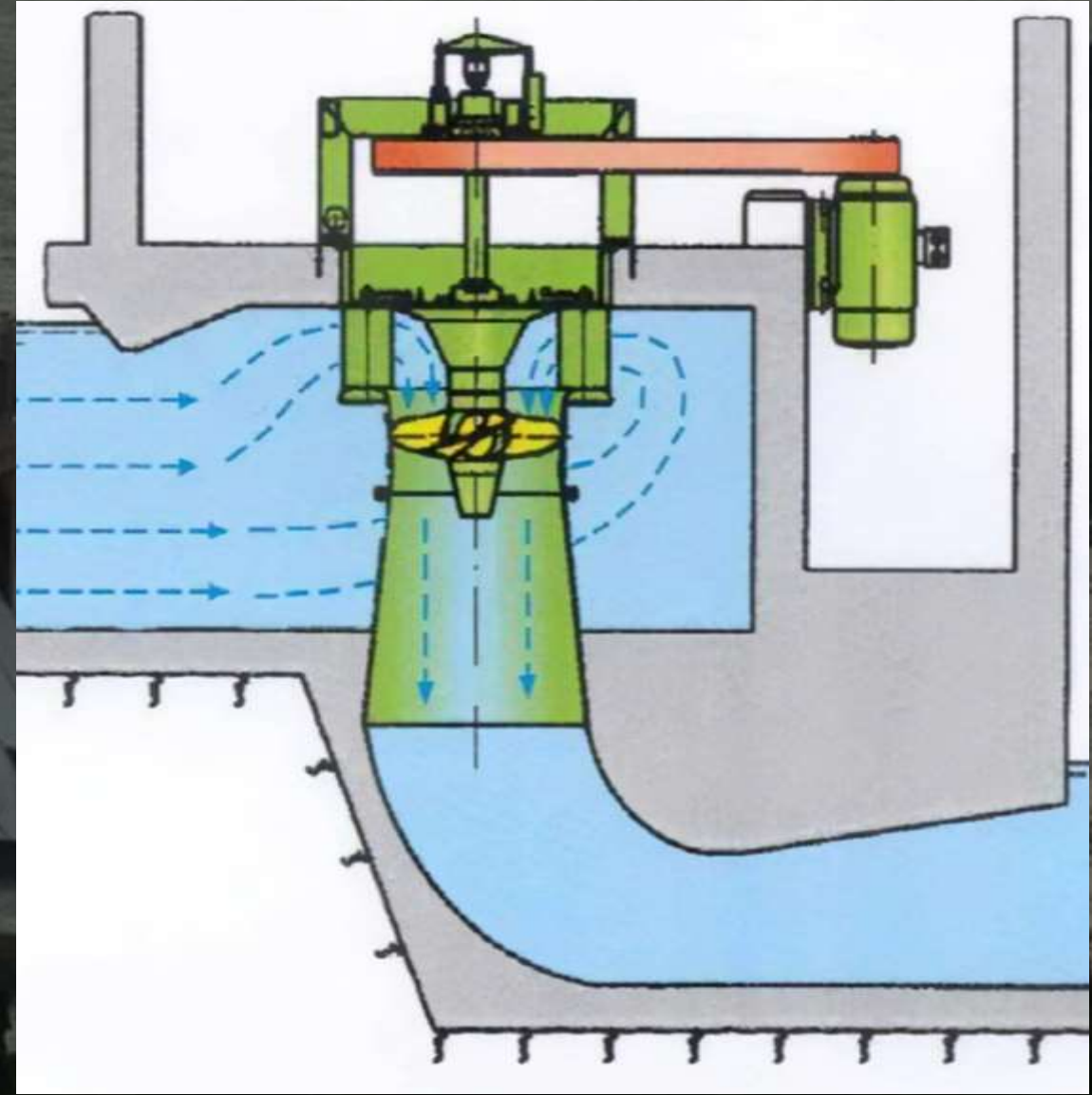
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Powerhouse and Water Conductor System

- Powerhouse: Surface type, 8 units, each 250 MW (fixed speed, reversible Francis turbines)
- Penstocks: 8 steel-lined, 7.5 m diameter, 582.44 m long (per unit)
- Tailrace Tunnel: 8 tunnels, 9 m diameter, 196 m long.
- Tailrace Channel: 874.45 m long, 120 m bed width, 6 m depth



Layout:

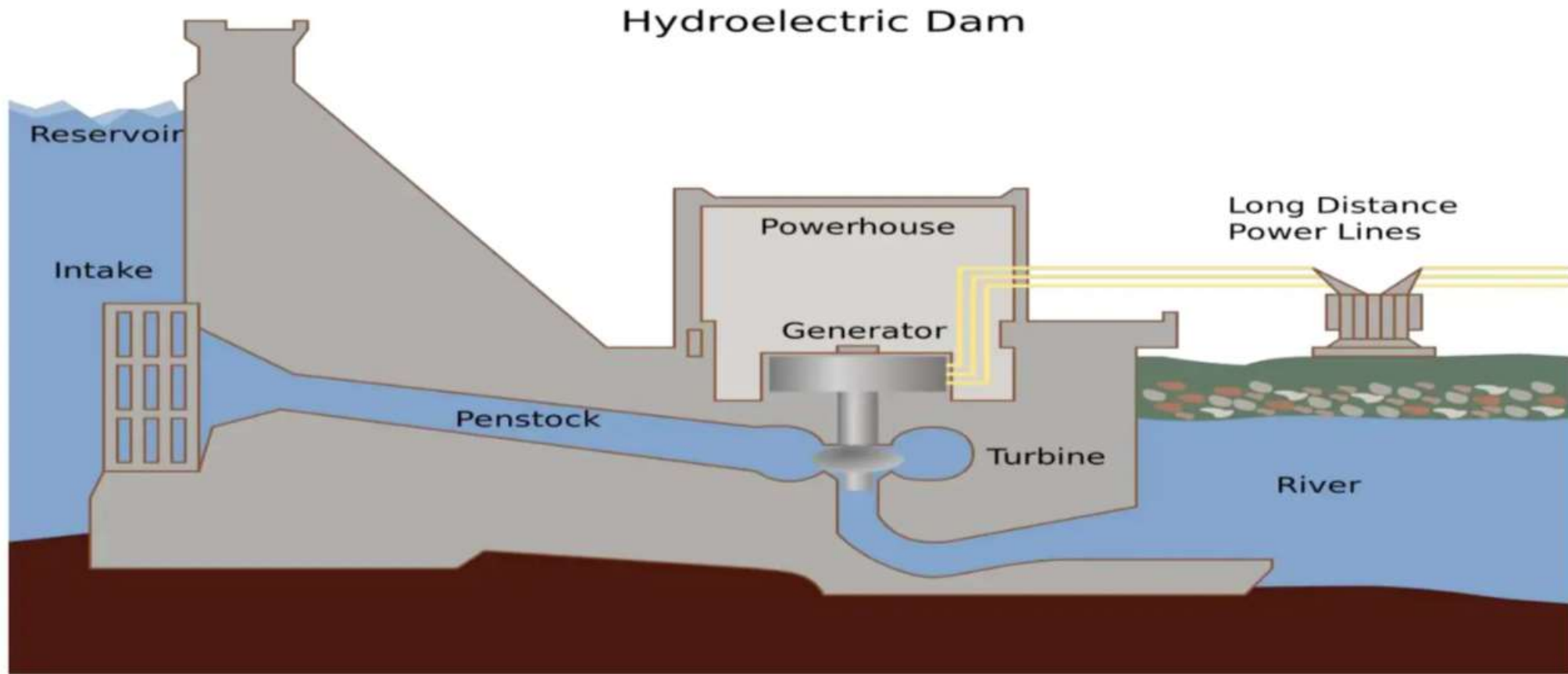


Fig: 1 Hydroelectric Power Plant Layout Diagram

Functions of Rana Pratap

Sagar

Primary objective of the dam was to produce hydroelectricity , and it generates 172MW of clean and renewable energy to support the locals.

Other purposes were:

- **Irrigation** which provides water to thousands of hectares of farmland, help in both rabi and kharif crop cycle.
- **Flood control**- it happens on regulating the flow of Chambal through the monsoons, or if stored in excess then releases in controlled manner to protect downstream towns .
- **Water supply**- provides drinking and industrial water

Also some minor reasons are like fishing , tourism and ground water recharge and ecological balance.

Reservoir OperationReservoir

Reservoir operation refers to **managing the storage and release of water** in the dam's reservoir to meet multiple objectives like

How the demand fluctuates the flow of water

- **Power demand:** More release during peak electricity demand
- **Irrigation needs:** Aligned with cropping seasons
- **Flood forecasting:** Emergency release if water level rises rapidly
- Uses **automated and manual gate systems** to control excess water

during floods while keeping structural safety intact.

Water is stored and released in a way that **turbines run at full efficiency**. Also peak electricity hours are matched with **high turbine output**.

In case of irrigation optimization releases are synchronised with agricultural cycles (Rabi and kharif seasons) to avoid both waterlogging and draught conditions in downstream.

Benefits of Optimized Operation

- Increases **power production** with minimal water use
- Ensures **timely irrigation** for farmers
- Reduces **flood damage**
- Supports **year-round water availability**

Maintains **ecological balance** downstream

Sources :-

- [Rana Pratap Sagar Dam - Bharatpedia](#)
- [Comprehensive EIA for Proposed RAPP unit 7 8 at Rawatbhata Rajasthan.pdf](#)
- [https://en.wikipedia.org/wiki/Rana Pratap Sagar Dam](https://en.wikipedia.org/wiki/Rana_Pratap_Sagar_Dam)
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- <https://www.power-technology.com/marketdata/power-plant-profile-rana-pratap-sagar-india/>

A large dam with water flowing over it, serving as a background for the text.

THANK YOU
